

Cancer Surveillance and Disparities: Looking at Prevention and Risk Factors in 2024

Introduction

Cancer is becoming a nationally growing issue, especially in the United States, and requires a deeper understanding of its incidence and associated risk factors across the state. Currently, cancer is one of the top five leading causes of death in the US, and it is continuing to grow. With changes to our environment or negative behavioral habits, the components leading to cancer in the US could be endless. To understand why it is going higher, a comprehensive surveillance program to track both incidence and survivorship could support the creation of effective preventive resources and slow the incidence rate. Observing through a broader lens of cancer would involve changing our interventions and updating older resources, such as revising the routine screening guidelines or evaluating risk factors associated with cancer survivorship.

This analysis uses the 2024 Behavioral Risk Factor Surveillance System (BRFSS) dataset to attain two primary objectives. First, to evaluate socioeconomic disparities in cancer prevention, by analyzing the relationship between educational level and colon cancer screenings with the Rao-Scott Chi-Square test. Second is to connect how specific demographic and behavioral risk factors like age, sex, smoking history, and Body Mass Index (BMI) are associated with overall cancer survivorship. This objective would involve a Quasibinomial Logistic Regression analysis.

Methods

This analysis uses data from the 2024 BRFSS, a national telephone survey with landline and cell that collects health behavior data from US adults. The 2024 sampling frame included 49 states, the District of Columbia, and participating US territories; however, Tennessee was removed from this dataset due to a lack of data collection. Additionally, the 2024 dataset had changes to recent state executive orders, which meant removing demographic questions such as transgender status and sexual orientation.

The dataset also has two potential errors, such as missing values and the accuracy of people's answers. First, the dataset forced missing values for certain variables. Second, there is a potential source of measurement error in people remembering their medical history, like misremembering the exact dates of their past colon cancer screenings, which can affect the validity of the final numbers.

Statistical analyses were conducted in R using the *survey* package to account for the complex sampling design, including the calculation of the CDC's complex weights. One of the tools the survey package has is the design object, which uses the primary sampling unit (`_PSU`), stratification (`_STSTR`), and final survey weights (`_LLCPWT`).

As for variance estimation, it was calculated using Taylor Series Linearization, the default method for handling complex survey samples.

Looking at the Missing data, the R software handles the missing values internally to keep variance estimates. Associations between education level and colon cancer screening were evaluated using a Rao-Scott Chi-Square test. To assess cancer survivorship, a quasi-binomial logistic regression was used to account for overdispersion in weighted survey data.

Results

Table 1. Colon Cancer Screening by Educational Level

Colon Cancer Screening by Educational Level					
Characteristic	Less than High School N = 29,296,197 ¹	High School Grad N = 71,810,489 ¹	Some College N = 77,728,162 ¹	College Grad N = 83,355,508 ¹	p-value ²
Screening_Status					<0.001
Up to Date	54%	67%	72%	76%	
Off Schedule	5.5%	6.5%	7.1%	6.4%	
Never Screened	40%	26%	21%	18%	
Unknown	17,675,422	45,681,164	44,052,717	46,387,989	
¹ %					
² Pearson's X ² : Rao & Scott adjustment					

The initial BRFSS dataset included 457,670 people. The analysis of colon cancer screening showed a strong, positive association between education and getting screened. About 54.2% of adults who did not complete high school reported being up to date on their colon cancer screenings, compared to about 67.3% of high school graduates, 72.0% of individuals with some college, and 76.0% of college graduates ($p < 0.001$).

To assess the impact of the survey design, the design effect ($deff = TRUE$) was calculated. For the college graduates, the design effect was 6.21, where the variance in our data was 6.2 times larger than if this had been a simple random sample. By having this effect applied, it proves that reasoning for using Taylor Series Linearization was needed.

Predictors of Cancer Survivorship (Odds Ratios)

Characteristic	OR	95% CI	p-value
age_group			
18-24	—	—	
25-34	1.78	1.26, 2.51	0.001
35-44	4.00	2.89, 5.55	<0.001
45-54	8.00	5.84, 11.0	<0.001
55-64	16.7	12.2, 22.9	<0.001
65+	34.2	25.1, 46.6	<0.001
Sex			
Male	—	—	
Female	1.15	1.10, 1.21	<0.001
smoking			
Smoker (100+ Cigs)	—	—	
Non-Smoker	0.79	0.75, 0.83	<0.001
BMI_C			
Underweight	—	—	
Normal Weight	0.91	0.76, 1.10	0.3
Overweight	0.91	0.75, 1.09	0.3
Obese	0.94	0.78, 1.13	0.5

Abbreviations: CI = Confidence Interval, OR = Odds Ratio

Table 2. Predictors of Cancer Survivorship (Odds Ratios)

Next, the focus was on overall cancer risk, and missing data on the regression covariates were retained for the unconditional analyses. The quasibinomial logistic regression was modeled using a subpopulation of 400,030 individuals. Age was the biggest factor, where the oldest group of people had 34.3 times the odds of reporting cancer compared to the youngest group ($p < 0.001$). To compare individuals by sex, the analysis had to be age-adjusted to obtain comparable odds ratios. After adjusting for age, smoking status, and BMI, females had 1.15 times the odds of reporting cancer compared to males ($p < 0.001$). Smoking status was also significantly associated with survivorship, where Non-smokers had 21% lower odds of reporting cancer compared to people who had smoked at least 100 cigarettes in their life ($p < 0.001$). In this specific model, BMI category did not make a statistically significant difference ($p > 0.05$).

Conclusions

The findings from the BRFSS tell a clear story that both socioeconomic status and behavioral risk factors play into the environment of cancer surveillance. The disparity in colon cancer screening rates based on people's educational level demonstrates a change in preventive care, such as outreach to lower education levels about screening programs. Additionally, a larger push for smoking cessation programs or a higher tax revenue on cigarettes could combat smoking risk habits within the US.

Narrowing down to these conclusions requires thorough steps and validity checks to make sure these statements are correct. Moreover, the analysis setup needed to represent the US population. The analysis was based on the survey package, where the `svydesign` function provides R with an instruction manual for interpreting the data. As mentioned, the design function consists of three pieces: PSU, stratification, and final survey weights. For the primary sampling unit (`_PSU`), this allows people to be grouped into clusters that share similar characteristics, such as income level or age. This would narrow down margins of error, or a person would be considered independent and uncorrelated, which would not be best for complex surveys like BRFSS. The second piece would be stratification (`_STSTR_`), which divides the entire population by state or specific geographic regions, ensuring

everyone has the same chance of being selected. This benefits groups such as rural areas and minority groups that would not be left out of the analysis. Finally, the survey weights (`_LLCPWT`) ensure that individuals who answer the survey are treated equally. In other words, people who answered the survey do not truly represent the actual US population. To reflect the US population, the weights serve as an equalizer, treating under-represented and over-represented groups as they would be if they had the same number of people.

The `svydesign` served as the blueprint for the data, but was not the only component of this analysis. Another component was the Design Effect (`deff = TRUE`), which is part of the `svydesign` function. It falls under the `svydesign` function because the calculated design effect compares complex surveys to a simple random sample. The calculated design effect ratio checks how much the complex `svydesign` function impacted the variance. For example, the analysis showed a design effect ratio of over 6.0, indicating that the variance in that population is 6 times greater than another statistical test would assume.

Both `svydesign` and the Design Effect were foundational components for this analysis, but needed more support to accommodate a large, complex survey. In a large complex survey, there was missing information, such as individuals not answering questions or not answering at all. The missing data were labeled as NAs and handled in the survey package, which retained the original 457,670 rows and still calculated the margins of error using only the people who answered.

In the statistical tests section, two tests were conducted. Firstly, the Rao-Scott Chi-Square test (`svychisq`) was used to test whether education levels were associated with screening for colon cancer. Compared to a simple Chi-Square test, such as the Pearson Chi-Square, the Rao-Scott Chi-Square test is intended for a complex design like the BRFSS, which includes clusters, strata, and weights. By applying the function (`svychisq`), the software looks at the `svydesign` function to see how clustered and weighted the data is, and it takes the standard Chi-Square test statistic and shrinks it down to the specific Design Effect. Using this Rao-Scott adjustment accounts for overlapping information in the BRFSS clusters. It shows their significant findings, such as who is more likely to get screened based on educational level, as well as logistic regression, where the Quasibinomial logistic regression (`svyglm`) model provides greater flexibility given the complexity of the clusters.

In contrast, the standard logistic regression would treat the data as independent subjects. Suppose the logistic regression treats the data as independent subjects, which corresponds to treating the data as a simple random sample. But as mentioned, BRFSS data is complex, with variance much higher than the model's assumptions, leading to overdispersion. Overdispersion would result in narrow confidence intervals and falsely significant p-values; moreover, the results would not be a true estimate of the data. With

the Quasibinomial model, the standard errors are scaled to match the complex clustered data in the BRFSS, resulting in accurate standard errors and confidence intervals. By using this model to evaluate age, sex, smoking history, and BMI at the same time, the analysis controls for within-cluster similarities, ensuring that the calculated odds ratios reflect the true predictors of cancer survivorship rather than false significance.

These findings have limitations, as this data reflects individuals at a specific time point during the BRFSS survey. This analysis is based on the associated rather than the establishing causation relationship between the variables. Moreover, this survey is conducted by phone or landline. Potential biases could include people remembering their answers and accurately reporting their medical history. Despite these limitations, the complex survey methodology provides a snapshot of cancer prevention and survivorship trends across the United States.

References

CDC. Cancer Data and Statistics. Cancer. Published May 17, 2024.
<https://www.cdc.gov/cancer/data/index.html>

Appendix A: R code

title: "Final Project Survey"

author: "Suzy Molina"

date: "2026-05-05"

output: html_document

```
` `` {r, include=FALSE}
```

```
library(haven)
```

```
library(dplyr)
```

```
library(survey)
```

```
library(gtsummary)
```

```
` `` `
```

```
` `` {r}
```

```
data <- read_sas("C:/Users/nmolin/OneDrive - University of Iowa/Documents/BIOS 6420  
Mary and Jake/llcp2024.sas7bdat")
```

```
options(survey.lonely.psu = "adjust") # controls single-PSU strata
```

```
design <- svydesign(  
  id = ~`_PSU`, # primary sampling unit  
  strata = ~`_STSTR`, # stratification  
  weights = ~`_LLCPWT`, # weight for landline/cell  
  data = data,  
  nest = TRUE  
)
```

```
# remember: at least two categorical responses
```

```
# two-way tables
```

```
# hypothesis test
```

```
##### Hypothesis Test #####
```

```
# Cancer screening vs Demographics
```

```
# Variable 1 (Response) _CRCREC3 - This is calculated variable for "Up to date on Color  
Cancer Screening"
```

```
# Variable 2 (Predictor) _EDUCAG - Computed level of education (check the levels)
```

```
##### Logistic Regression (binary) #####
```

```
### Cancer Survivor ship Status ###
```

```
# Response Variable (Binary): CHCSCNC1 - "Ever told you had skin cancer?"
```

```
# CHCOCON1 - "Ever told you had any other types of cancer?"
```

```
# Covariates: _AGE_G - Imputed age groups
```

```
# _SEX - calculated sex
```

```
# SMOKE100 - Have you smoked at least 100 cigarettes in your entire life?
```

```
#_BMI5CAT - Computed Body Mass Index categories.
```

```
design <- update(design,
```

```
  cancer_binary = ifelse(CHCOCNC1 == 1, 1,
```

```
  ifelse(CHCOCNC1 == 2, 0, NA)))
```

```
# SMOKE100 AND _SEX ARE 1S AND 2S
```

```
design <- update(design,
```

```
  age_group = factor(`_AGE_G`,
```

```
    levels = c(1, 2, 3, 4, 5, 6),
```

```
    labels = c("18-24", "25-34", "35-44", "45-54", "55-64", "65+")),
```

```
  smoking = factor(SMOKE100,
```

```
    levels = c(1, 2),
```

```
    labels = c("Smoker (100+ Cigs)", "Non-Smoker")),
```

```
  BMI_C = factor(`_BMI5CAT`,
```

```
    levels = c(1, 2, 3, 4),
```

```
    labels = c("Underweight", "Normal Weight", "Overweight", "Obese")),
```



```
Sex = factor(`_SEX`,  
             levels = c(1, 2),  
             labels = c("Male", "Female"))))
```

```
cancer_model <- svyglm(cancer_binary ~ age_group +  
                      Sex + smoking + BMI_C,  
                      design = design,  
                      family = quasibinomial())  
summary(cancer_model)
```

```
# Odds Ratio of the model  
exp(coef(cancer_model))
```

```
##### TWO-WAY TABLE (EDUCATION VS. COLON CANCER SCREENING)  
#####
```

```
design <- update(design,  
               Education_Level = factor(`_EDUCAG`,  
                                         levels = c(1, 2, 3, 4),  
                                         labels = c("Less than High School", "High School Grad", "Some  
College", "College Grad")),  
               Screening_Status = factor(`_CRCREC3`,  
                                         levels = c(1, 2, 3),  
                                         labels = c("Up to Date", "Off Schedule", "Never Screened")))
```

```
#'deff = TRUE' -> the design effect as in lecture 6
```

```
screening_table <- svytable(~`_CRCREC3` + `_EDUCAG`,  
                           design = design)
```

```
screening_deff <- svyby(~as.factor(`_CRCREC3`), ~as.factor(`_EDUCAG`),  
                      design = design,  
                      FUN = svymean,  
                      deff = TRUE,  
                      na.rm = TRUE)
```

```
print(screening_deff)
```

```
# This is Rao-Scott Chi-Square test for significance
```

```
svychisq(~`_CRCREC3` + `_EDUCAG`, design = design)
```

```
` ``
```

```
` `` {r}
```

```
##### Logistic Regression Table #####
```

```
r_table <- tbl_regression(cancer_model, exponentiate = TRUE) %>%
```

```
bold_p(t = 0.05) %>%
```

```
modify_caption("***Table 1. Predictors of Cancer Survivorship (Odds Ratios)**")
```

r_table

Two-Way Table

```
screening_summary_table <- design %>%
```

```
  tbl_svsummary(
```

```
    by = Education_Level,
```

```
    include = c(Screening_Status),
```

```
    statistic = list(all_categorical() ~ "{p}%")
```

```
  ) %>%
```

```
  add_p() %>%
```

```
  modify_caption("**Table 2. Colon Cancer Screening by Educational Level**")
```

screening_summary_table

...

Appendix B: AI Assistance

AI Assistance

Research and Information

☒ I conducted all research using traditional methods

☐ I utilized AI-powered search tools to find relevant information

☐ I used AI-summarized articles but drew my own conclusions

☐ I assessed AI-generated reviews of sources and expanded upon them

Planning and Organization

☐ I organized and structured my report on my own

☒ I used generative AI to refine my own outline using AI-suggested improvements

☐ I started with an AI-generated outline and developed it with my own insights

☐ I collaborated with AI throughout the project, taking suggestions and improving them in ways that helped strengthen my ideas

Content Development

☐ I wrote all content without AI assistance

☐ I used AI for proofreading and minor language improvements

☐ I expanded on AI-generated paragraphs with my own knowledge and creativity

☐ I guided AI content generation and substantially edited the output

Statistical Coding

☐ I wrote all code without AI assistance

☐ I used AI for software language improvements

☐ I expanded on AI-generated coding suggestions with my own knowledge and creativity

☐ I guided AI content generation and substantially edited the code

Editing and Refinement

☐ I edited and refined my work independently

☐ I considered AI style suggestions and made informed choices

☐ I critically evaluated AI-suggested rewrites and selectively implemented them

☐ I collaborated with AI for comprehensive edits, but made sure to maintain my voice and intent